WORK STUDY

- Today's competitive market industry must be aware of latest developments for new technology.

1. By improving the process of manufacture
2. By improving the operation of existing facilities(Plant and Human)

DEFINITION OF WORK STUDY

- Study of human work in all aspects in order to increase the effectiveness with which work is done.
- Work study aimed to achieving higher productivity
- By finding the most efficient use of available resource to create output.

OBJECTIVES OF WORK STUDY

- Most economical way for work
- To simplify and standardise the methods, materials and tools.
- Time required for a qualified worker
- To plan training program for the worker for new method.

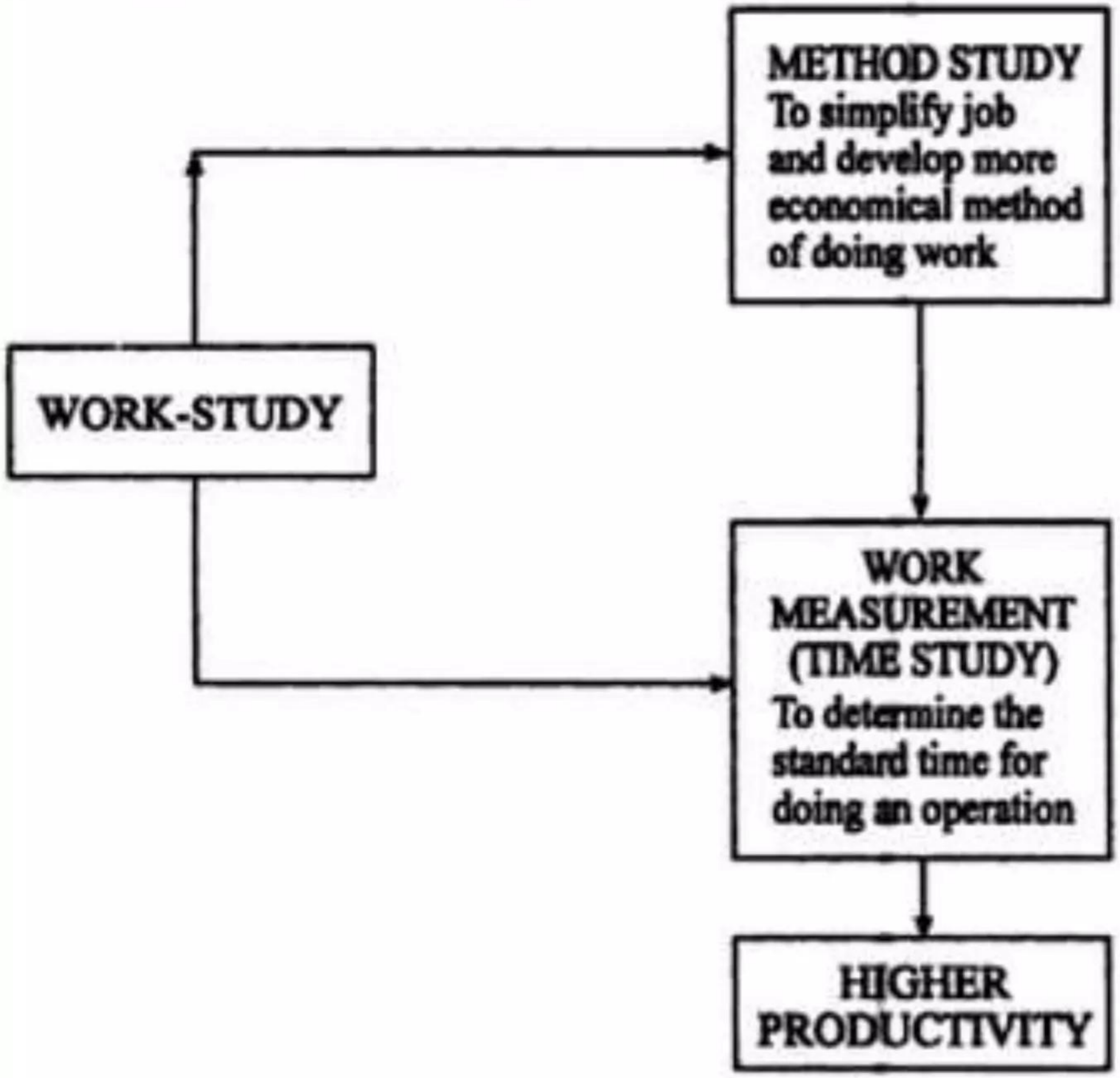
PURPOSE OF WORK STUD

- Raising productivity
- Increase productivity without capital expenditure
- It helps to eliminate / reduce waste
- Better quality of product at reasonable price

Work study procedure (13 ma

1. Select (job or process selected)
2. Record (record by various techniques)
3. Examine (asking questions who,when)
4. Develop (Economical method)
5. Measure (amount of work involved)
6. Define (new method)
7. Install (new method standard pratice)
8. Maintain (proper control the above pratice)

WORK STUDY COMPONENTS



WORK STUDY

- ***Method study***
- *It is the systematic recording & critical examination of existing and proposed ways of doing work, as a means of developing and applying easier and more effective methods and reducing cost*

- ***Work measurement / Time study***
- *It is the application of techniques designed to establish the time for a qualified worker to carry out a specified job at a defined level of performance*

METHOD STUDY

Objectives

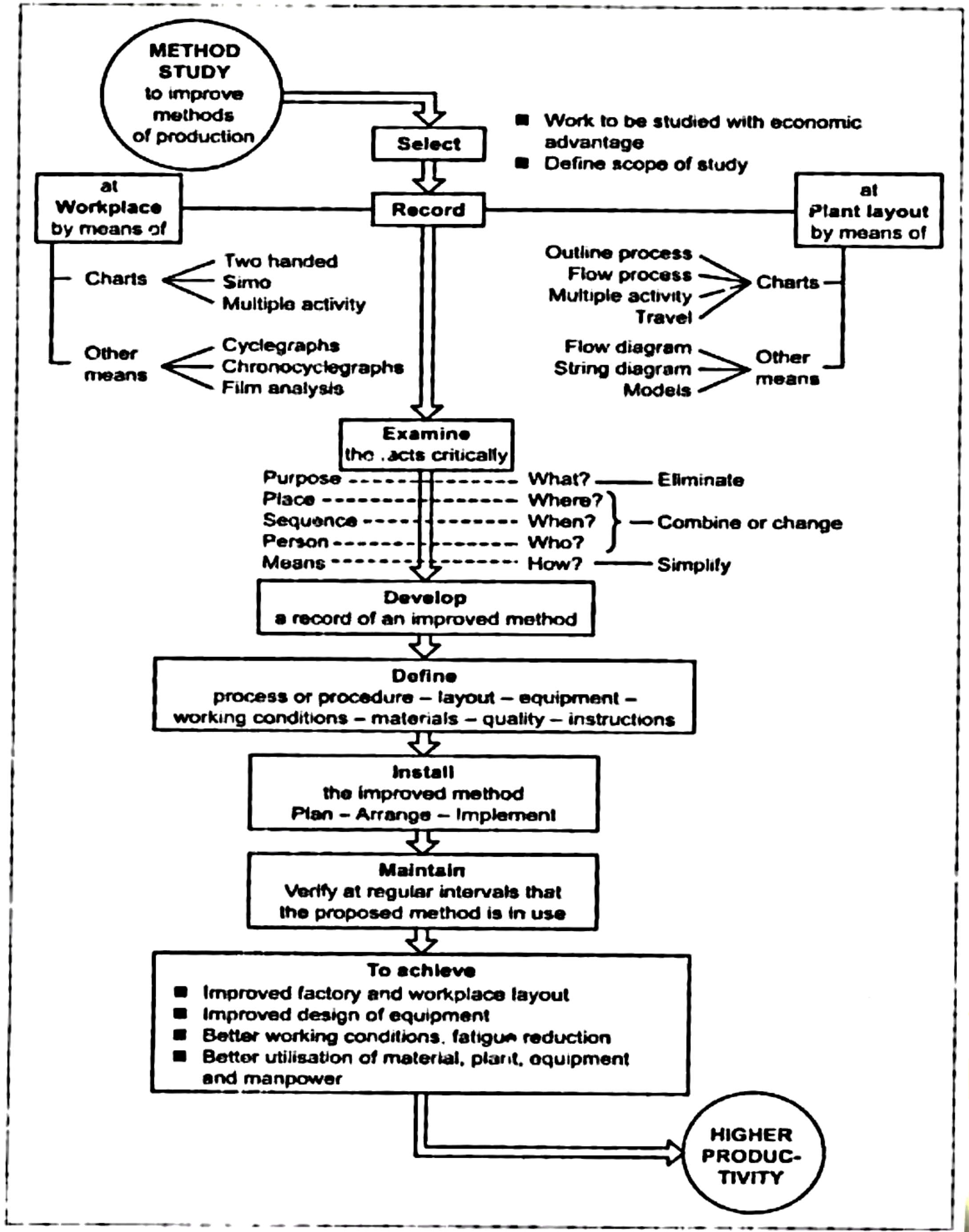
- *Critical examination of facts*
- *Develop best possible solution*
- *Eliminate unnecessary operations*
- *Add value & Avoid delays*
- *Optimize 3M*

Objectives of Method study

- To improve process and procedures
- To improve design plant and equipment
- Improve plant layout
- Improve men , material.
- Standardise methods
- Improve safety methods

Basic procedure for method study (13 Mark)

1. Select (job or process selected)
2. Record (record by various techniques)
3. Examine (asking questions who,when)
4. Develop (Economical method)
5. Define (new method)
6. Install (new method standard pratice)
7. Maintain (proper control the above pratice)


METHOD STUDY to improve methods of production
Select
Work to be studied with economic advantage
Define scope of study
at Workplace by means of
Record
at Plant layout by means of
Charts
Two handed
Simo
Multiple activity
Other means
Cyclegraphs
Chronocyclegraphs
Film analysis
Outline process
Flow process
Multiple activity
Travel
Charts
Flow diagram
String diagram
Models
Other means
Examine the facts critically
Purpose
Place
Sequence
Person
Means
What?
Where?
When?
Who?
How?
Eliminate
Combine or change
Simplify
Develop a record of an improved method
Define process or procedure – layout – equipment – working conditions – materials – quality – instructions
Install the improved method Plan – Arrange – Implement
Maintain Verify at regular intervals that the proposed method is in use
To achieve
Improved factory and workplace layout
Improved design of equipment
Better working conditions, fatigue reduction
Better utilisation of material, plant, equipment and manpower
HIGHER PRODUCTIVITY

1) SELECTION OF JOB FOR METHOD STUDIES

- 1) Economic considerations:
  - Waste of time for long investigation
  - Operations involved excess man power
  - By tracking materials
  - Repetative work
  - Operations having poor utilisation of resources
- 2) Techinical consideration:
- To ensure adequate technology is available to carry out the study successfully.
- 3) Human reactions:
- All concern persons accept the method study for improve.

- <u>2) RECORD</u>
- After selection the job record the Relevant facts of existing methods.
- RECORDONG METHODS
  - CHARTS
    - Outline process chart
    - Flow process chart
    - Multiple activity chart wth time scale
    - Simo chart
  - CHARTS AND DIAGRAMS
    - Flow diagram
    - String diagram

- 3) EXAMINE
- Relevant informations and procedures are recorder next stem examine that.
- CRITICAL QUESTIONS
  1. PRIMARY QUESTIONS
  2. SECONDARY QUESTIONS

4) DEVELOP

Evaluate successfully the four step formula are

1. Eliminate all un necessary operations
2. Combine two or more operations
3. Sequence of various activities
4. Simplify necessary operations

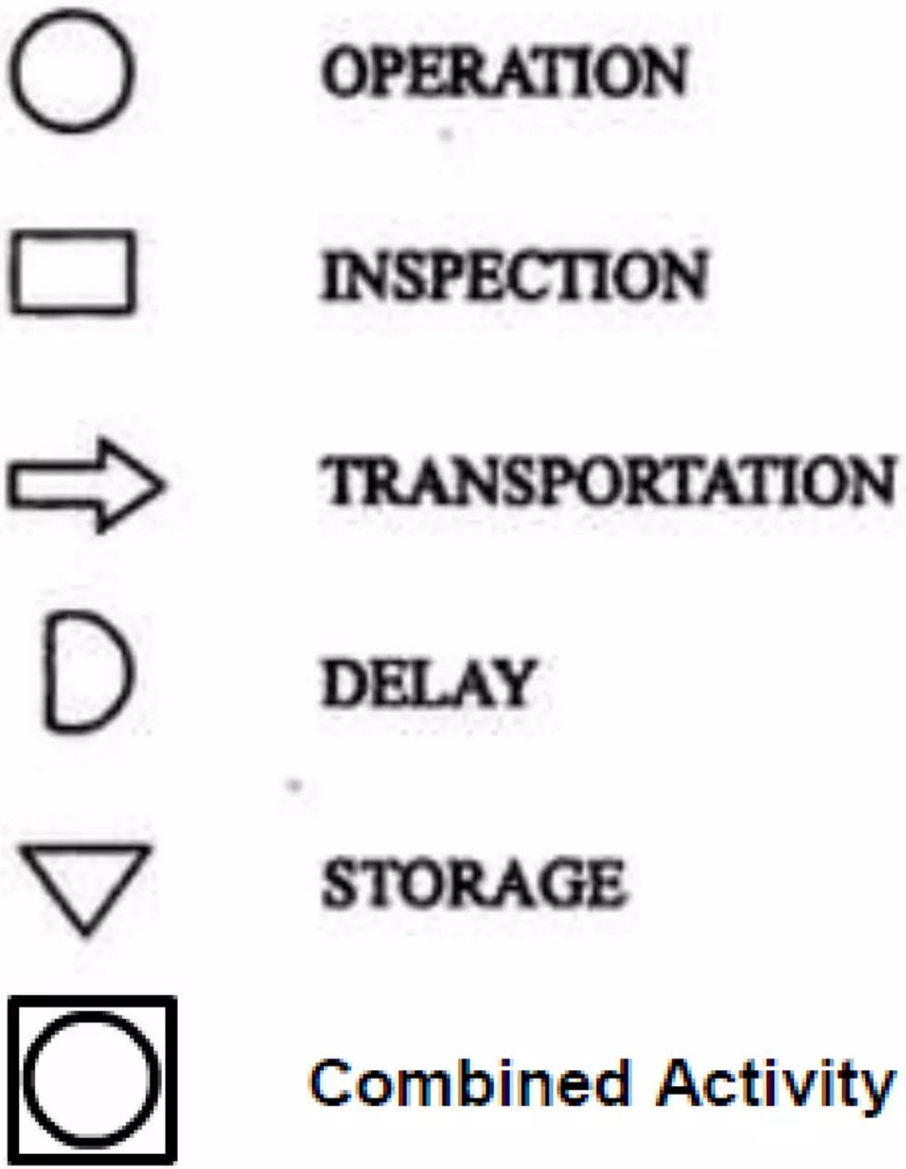
- 5) DEFINE NEW METHOD
  - By analysing the above methods We Define newer method by agreeing the various workers and management.
- 6) INSTALL NEW METHOD
  - Install new method
  - Train the workers for the new method
  - The proposed method was accepted by supervisors, etc.

  7) MAINTAIN THE PROPOSED METHOD
  - Adopting various monitoring techniques
  - Data collection and interruption
  - Feed back from evaluated reports

RECORDING TECHNIQES (13 mark)

1. CHARTS
   1. METHOD STUDY
   2. PROCESS CHARTS
      1. OUTLINE PROCESS CHARTS
      2. FLOW PROCESS CHARTS
      3. TWO HANDED PROCESS CHARTS
2. DIAGRAMS
   1. FLOW DIAGRAM
   2. STRING DIAGRAM
   3. TRAVEL CHART
   4. CYCLE GRAPH

Method Study Symbols



PROCESS CHART

- 1) OUTLINE PROCESS CHART
  - It is otherwise called as operation process chart
  - Overall view of the whole process
  - INPUT- OPERATION- OUTPUT

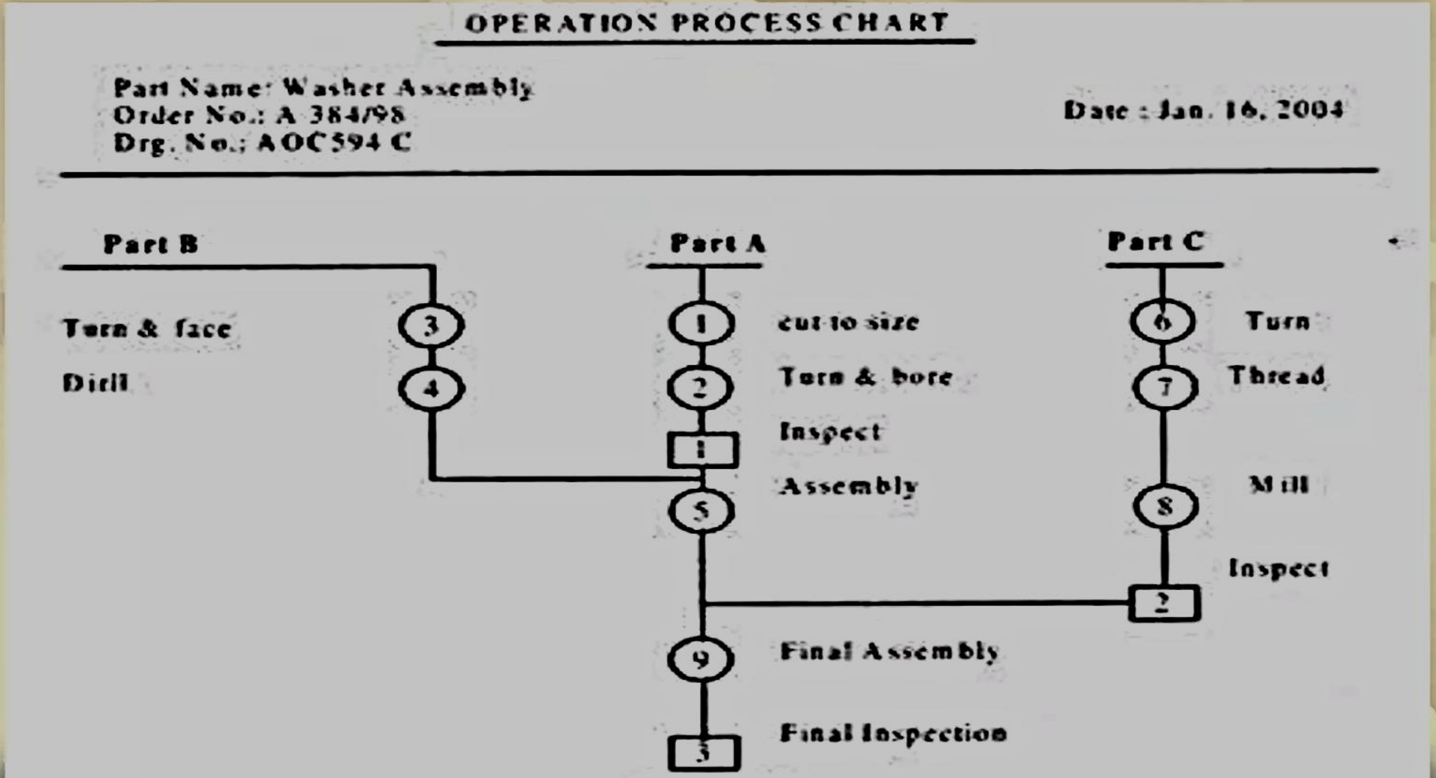
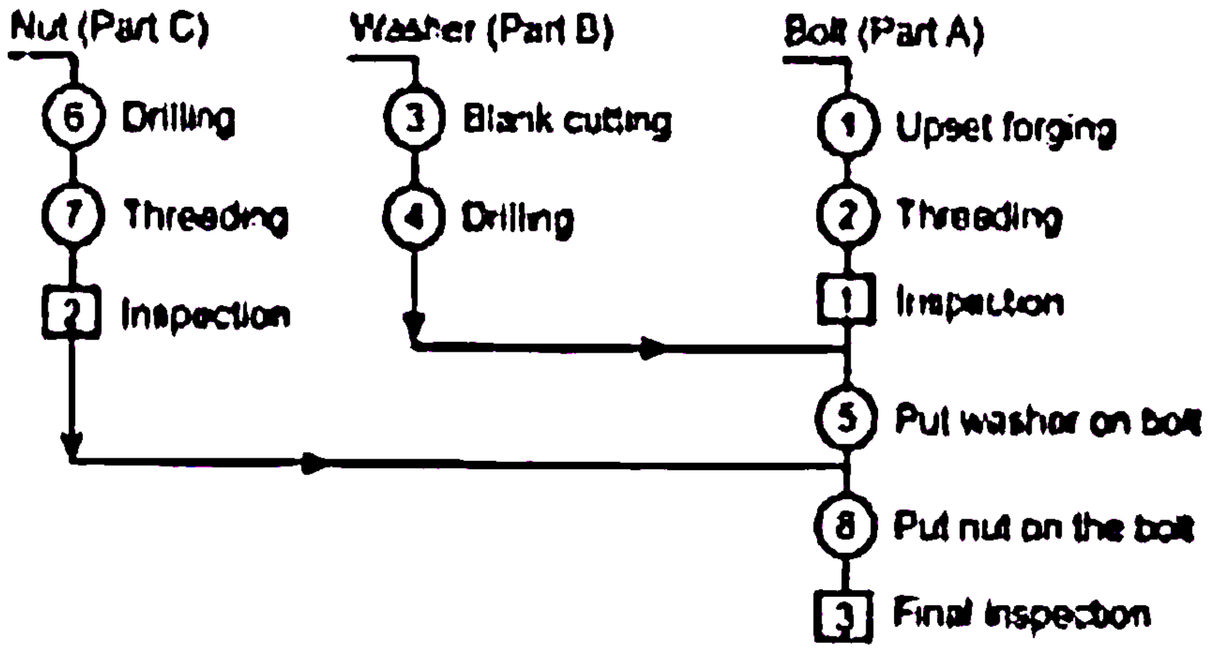


Fig. 18.1

Operation Process Chart	Name of the firm:
Name & description of assembly: Bolt and washer assembly	Sketch:

Charted by:	Date:	Existing ✓	Improved

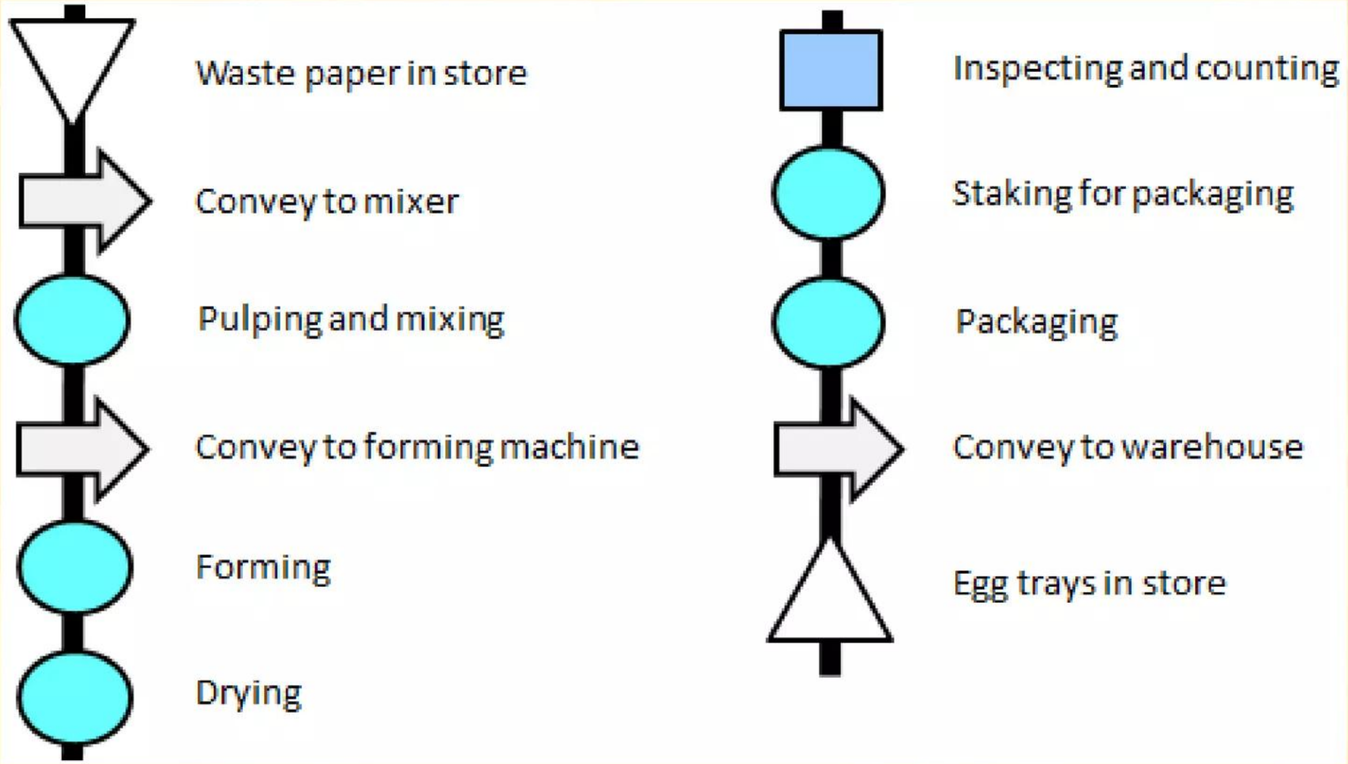


Summary	Existing	Improved	Saving
No. of operations	8		
No. of inspections	3		

2) FLOW PROCESS CHART

- It includes graphical representation of ALL OPERATIONS, INSPECTIONS, DELAYS, STORAGES etc.,
- DEF: Sequence of flow of product or procedure
  - TYPES OF FLOW CHART
    - Flow process chart (Man type)
    - Flow process chart (Material type)
    - Flow process chart (Equipment type)

FLOW PROCESS CHART EXAMPLE



FLOW PROCESS CHART		MAN/MATERIAL/EQUIPMENT TYPE			
Chart No.1 Sheet No.1 OF1		SUMMARY			
Subject charted: Welding of two plates		Activity	Present	Proposed	Saving
Chart Begins:	Man in the workplace	Operation ○	4		
Chart Ends:	Man places the welded plates in storage bin	Transport ⇨	5		
		Delay D	0		
		Inspection □	1		
		Storage ▽	1		
Method: Present		Distance (m)	7		
Charted By:		Time (man-machine)	—		
Approved By:	**Date:**	Total cost	—		

Description	Symbol ○	⇨	D	□	▽	Distance (m)	Time (min)	Remarks
Operator goes to collect unwelded plates		●				3		
Collects unwelded plates	●							
Goes to almirah to collect welding rods, storage bin, *etc.*		●				1		
Collects welding material	●							
Goes to work place		●				2.5		
Keeps the plates and other material at work place	●							
Goes to switch on electric supply		●				0.25		
Comes back to work place		●				0.25		
Does the welding	●							
Inspects the welded plates				●				
Keeps the welded plates in storage bin					●			
Total	4	5	0	1	1	7		

Fig. 1.6. Flow process chart – Man type (for welding of two plates)

FLOW PROCESS CHART	MAN/MATERIAL/EQUIPMENT TYPE			
Chart No.1 Sheet No.1 OF1	SUMMARY			
Subject charted: Gear **Chart Begins:** Manufacturing **Chart Ends:** Material in stores finished gear in storage bin	Activity	Present	Proposed	Saving
	Operation ○	4		
	Transport ⇨	5		
	Delay D	2		
	Inspection □	2		
	Storage ▽	2		
Method: Present	Distance (m)	55		
Charted By: **Approved By:** **Date:**	Time (man-machine)	29		
	Total cost	—		

Description	Symbol ○	⇨	D	□	▽	Distance (m)	Time (min)	Remarks
Bar from stores					•			
Sent to cutting machine		•				10		
Cut to size	•						2.5	
Sent to lathe		•				30		
Awaiting			•				1.5	
Facing, drilling and reaming	•						2	
Sent to gear hobbing machine		•				4		
Awaiting			•				1	
Cut the gear teeth	•						5	
Inspected				•			1	
Sent to heat treatment department		•				10		
Hardening	•						15	
Inspected				•			1	
Moved to stores		•				15		
Stored for reissue				•				
Total	4	5	2	2	2	55	29	

Fig. 1.7 Flow process chart—Material type (for gear milling)

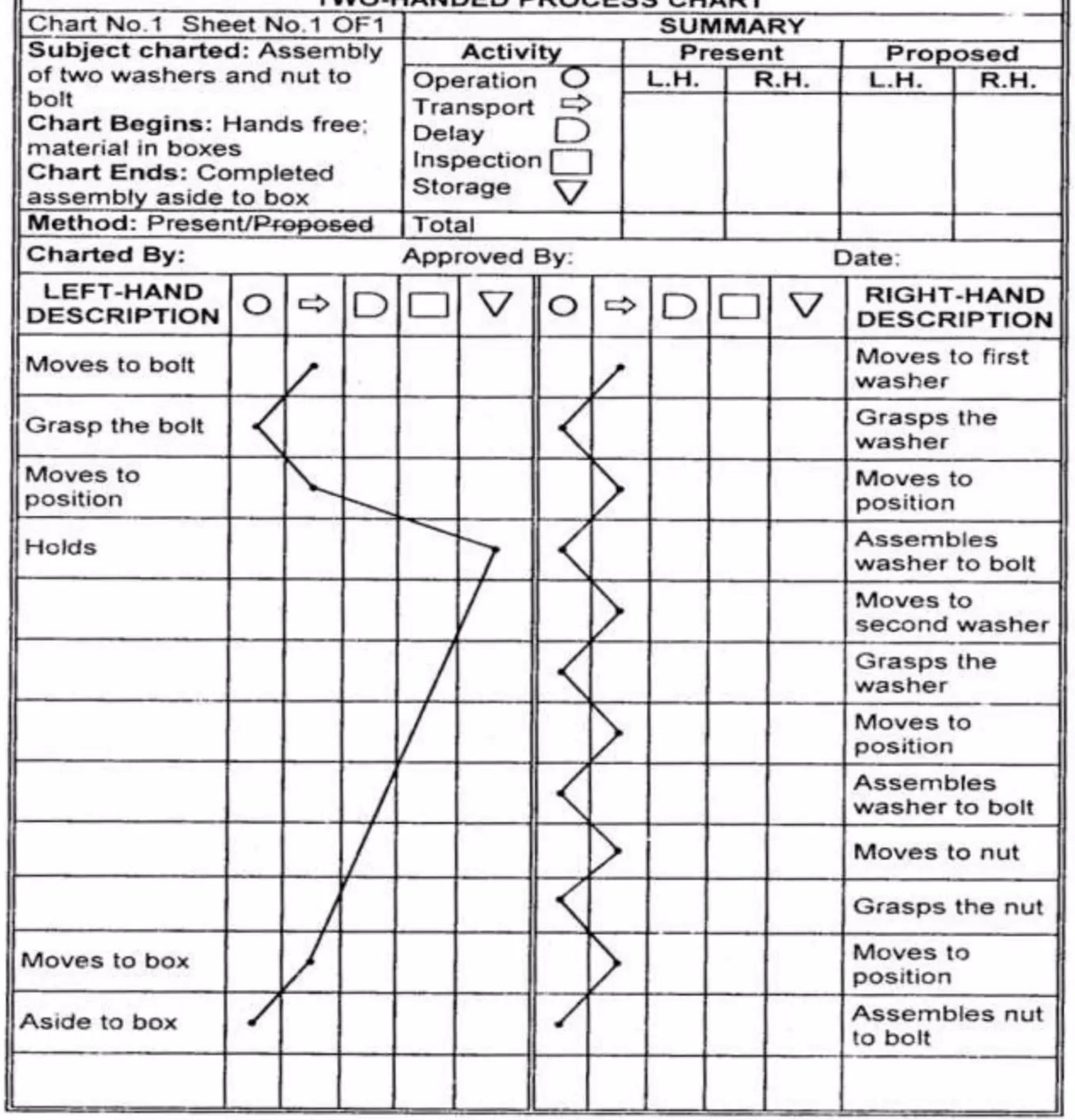
Recording Techniques - Charts

✓ *Flow process chart usefulness:*

- *Reduce travel distance of man/material*
- *Avoid waiting time & unnecessary delays*
- *Reduce cycle time by combining or eliminating operations*
- *Fix up the sequence of operations*
- *Relocate the inspection stages*

TWO HANDED PROCESS CHART

- Other wise called as left and right handed process chart
- Activities of the worker's hand are recorded
- To eliminate the unwanted motion to minimum
  - ADV:
    - Balance the motion and reduce fatigue
    - To eliminate and reduce motions
    - Train new operations in ideal method



TWO-HANDED PROCESS CHART

Chart No.1 Sheet No.1 OF1	SUMMARY				
Subject charted: Assembly of two washers and nut to bolt **Chart Begins:** Hands free; material in boxes **Chart Ends:** Completed assembly aside to box	**Activity**	**Present**		**Proposed**	
		L.H.	R.H.	L.H.	R.H.
	Operation ○				
	Transport ⇨				
	Delay D				
	Inspection □				
	Storage ▽				
Method: Present/~~Proposed~~	Total				

Charted By: Approved By: Date:

LEFT-HAND DESCRIPTION	○	⇨	D	□	▽	○	⇨	D	□	▽	RIGHT-HAND DESCRIPTION
Moves to bolt											Moves to first washer
Grasp the bolt											Grasps the washer
Moves to position											Moves to position
Holds											Assembles washer to bolt
											Moves to second washer
											Grasps the washer
											Moves to position
											Assembles washer to bolt
											Moves to nut
											Grasps the nut
Moves to box											Moves to position
Aside to box											Assembles nut to bolt

Fig. 1.8. Two-handed process chart (for assembling two washers

Flow Process Chart Job : Requisition of petty cash	Analyst ABC	Page 1 of 2					
Details of method			**Operation**	**Movement**	**Inspection**	**Delay**	**Storage**	**Distance**

Details of method	Operation	Movement	Inspection	Delay	Storage	Distance
Requisition made out by department head	●	⇨	□	D	▽	
Put in "pick-up" flag	○	⇨	□	◗	▽	
To accounting department	○	➡	□	D	▽	10 m
Account and signature verified	○	⇨	■	D	▽	
Amount approved by treasurer	●	⇨	□	D	▽	
Amount counted by cashier	●	⇨	□	D	▽	
Amount recorded by bookkeeper	●	⇨	□	D	▽	
Petty cash sealed in envelope	●	⇨	□	D	▽	5 m
Petty cash carried to department	○	➡	□	D	▽	
Petty cash checked against requisition	○	⇨	■	D	▽	
Receipt signed	●	⇨	□	D	▽	
Petty cash stored in a box	○	⇨	□	D	▼	
	○	⇨	□	D	▽	
	○	⇨	□	D	▽	
	○	⇨	□	D	▽	
	○	⇨	□	D	▽	

	Summary	**Distance**
Operations	6	
Inspections	2	
Transport	2	15 m
Delays	1	
Total	11	

CHARTS USING TIME SCALE

- 1) MULTIPLE ACTIVITY CHART:
  - It is a chart more than one subject(men, machine, item) are recorded on a common time scale
- USES OF MULTIPLE ACTIVITY CHART:
  - To analyses ideal time of men and machine.
  - Determine number of machines handle by an operator
- IMPORTANT TYPES OF MULTI ACTIVITY CHART:
  1. Man machine activity(1 operator, 1 machine)
  2. Multi man activity(many operators, 1 machine)
  3. Man-multi machine activity(1 operator, many machine)
  4. Multi machine activity chart (group of operators ,central machine)

- 2) <u>SIMO chart (SIMULTANEOUS MOTION CYCLE CHART)</u>
  - Record the activities, of two hands or other parts.
  - SIMO chart constructed from Data collected motion flim analysis
- <u>THERBLINGS</u>
  - Therblings are the symbols used to denote various activities and movements done for different purpose.

THERBLINGS


Search
Find
Select
Grasp
Hold
Transport Loaded
Transport Empty
Position
Assemble
Use
Disassemble
Inspect
Preposition
Release Load
Unavoidable Delay
Avoidable Delay
Plan
Rest

SIMO CHART

DRAWING No. AND NAME: 1 Bolt Washer Assembly

OPERATION: Assemble 2 washers on bolt

OP. No. A25

OPERATIVE

Film No.

Chart No.

Sheet No.

Charted By.

Date:

Wink Counter Reading	Left Hand Description	Therblig	Time	Time in 2000/min	Time	Therblig	Right Hand Description
				0			
120	Reaches for bolt	TE	12		12	TE	Reaches for bolt
	Selects & grasps bolt	STG	8	20		STG	Selects & grasps bolt
140	Carries bolt to fixture	TL	16	40	16	TL	Carries bolt to fixture
160	Position bolt into fixture	P	20	60	20	P	Position bolt into fixture
180	Reaches for steel washer	TE	12		12	TE	Reaches for steel washer
	Selects & grasps washer	STG	8		8	STG	Selects & grasps washer
200	Carries washer to bolt	TL	16	80	16	TL	Carries washer to bolt
	Positions washer into bolt	P	8	100	8	P	Positions washer into bolt
220	Reaches for lock washer	TE	12		12	TE	Reaches for lock washer
	Selects grasps block washer	STG	8	120	8	STG	Selects & grasps washer
240	Carries lock washer to bolt	TL	16	140	16	TL	Carries lock washer to bolt
	Position L washer to bolt	P	8		8	P	Position lock washer to
260	Withdraw assembly from fixture	DA	20	160	20	DA	bolt withdraw assembly from fixture
280	Carry assembly to bin	TL	16	180	16	TL	Carry assembly to bin
300	Release assembly	RL	8	200	8	RL	Release assembly

Fig. 1.10. SIMO chart for bolt and washer assembly

DIAGRAMS

- 1) FLOW DIAGRAMS:
  - Plan drawing in a working area, showing a location of various activities by symbols.
  - CONSTRUCTION PROCEDURE:
    - Draw the work area
    - Mark the positions of tools, benches
    - Trace the path
    - Indicate the direction of movement
    - Different colours for the different type of movement

FLOW DIAGRAM

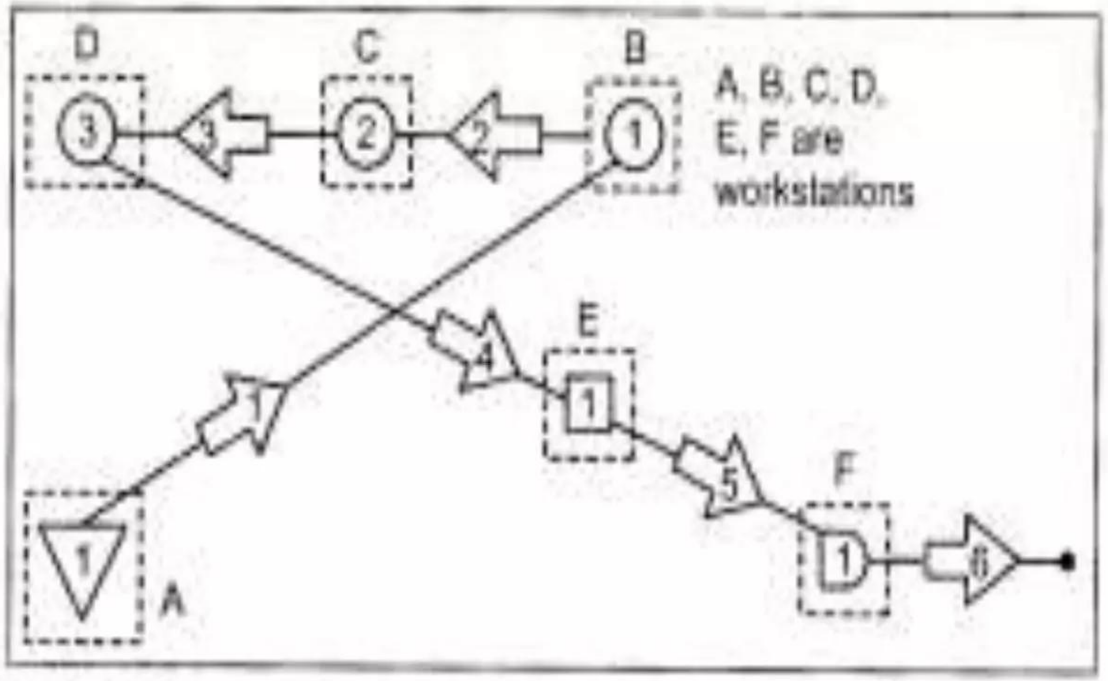


Fig. 1.11. Flow diagram


Patient
Assessment/Investigation process
Diagnosis made
Diagnosis excluded
Tx works
Risk avoided

STRING DIAGRAM

- It is a scale plan or model THREAD is used to trace and measure the path of materials , machine in define path.

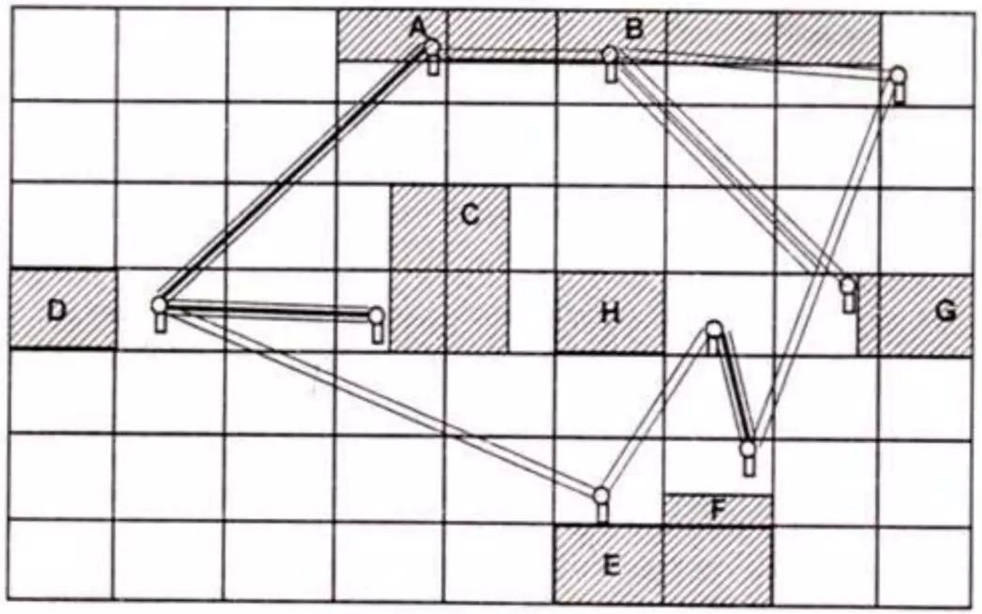


Fig. 4.7. String Diagram.

TRAVEL CHART

- A travel chart is a tabular record for presenting quantitative data about the movement of workers , materials or equipment between any number of places over any given period of time.

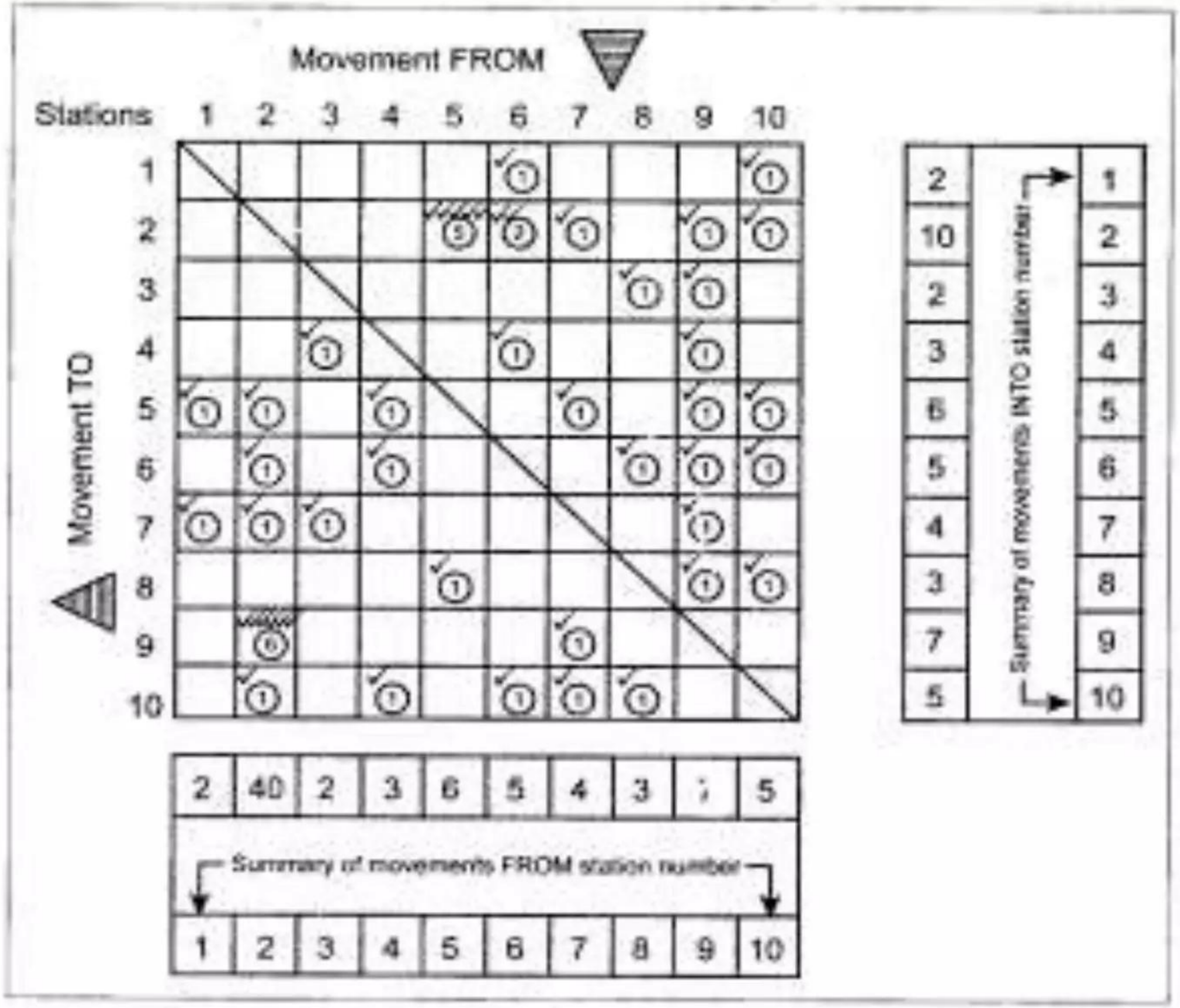


Fig. 1.13. Travel chart

CYCLE GRAPH

- The techniques are developed by Gilberth.
- It is the record of path movement , Usually traced by continuous source of light on photograph.
- For these small electric bulb is attached to the hand or finger, the motion can be recorded.

CHRONOCYCLE graph is a special form of cycle graph in which light source is suitable interrupted so thet path appears peer- dot shaped.

(a) Cycle graph.

(b) Chronocycle graph.

Fig. 4.11.

Recording Techniques - Diagrams

- ***Cycle graph & Chronocycle graph***

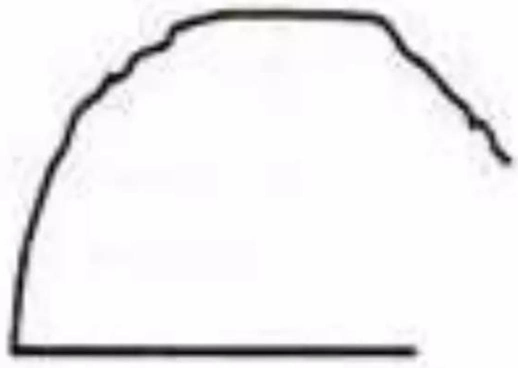
- *Both **records the motion path** of an operator & requires filming equipment*

- *Movements which are very fast and very **difficult for the human eye** to trace are traced by these techniques*

RECORDING TECHNIQUES - DIAGRAMS

➢ ***Cycle graph***

➢ *Indicates a permanent record of the motion pattern employed in the form of a closed loop of **continuous line**.*

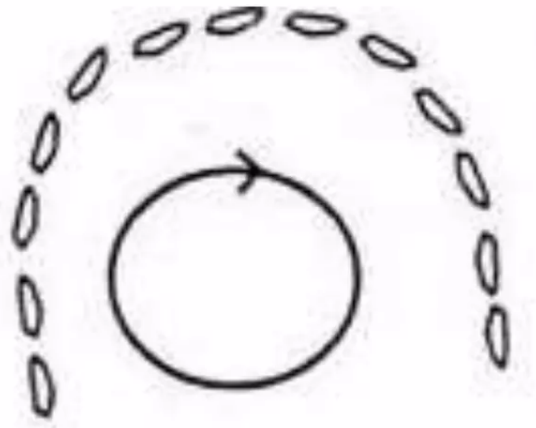


Cycle graph.

➢ *It **does not** indicate the direction or speed of motion.*

Recording Techniques - Diagrams

- ***Chronocycle graph***
- *Indicates* ***short dashes of line spaced*** *in proportion to the speed of the body member photographed*



Chronocycle graph.

- ***Wide spacing*** *would represent fast moves while* ***close spacing*** *would represent slow moves*

- ***Jumbling of dots*** *at one point would indicate fumbling or hesitation of the body member*

- *Used to study the* ***motion pattern*** *as well as to* ***compute velocity, acceleration and retardation*** *experienced by the body member at different locations.*

PRINCIPLES OF MOTION ECONOMY

- These principles are developed from improved methods
- These principles are groped in to THREE headings
    1. Use of human body
    2. Arrangement of workplace
    3. Design of tools and equipment.

Principles with the use of HUMAN BODY

- Two hands should began and end their movements in same time.
- Two hands should not be ideal at same time
- Hand motions should symmetrically opposite direction
- Momentum should be help of the worker during work
- Countineous curved movements preferred to straight line motions during sudden changes in direction.
- Free swinging movements are faster for controlling movements
- Work should be arranged to eye comfort for easier work

PRINCIPLES OF THE ARRANGEMENT OF WORKPLACE

- Fixed station to provide all tools
- Pre position of tools to reduce searching
- Scrap bins are closer to the working area
- Tools should be arranged permit for the best sequence
- Ejectors should provide whenever possible for disposal purpose.

PRINCIPLES OF TOOLS AND EQUIPMENTS

- Equipments are fixed by using jigs and fixtures.
- Two or more tools should be combined whenever possible
- Load should be distributed accordingly for the type writing purpose
- Handles such as large screw drivers are designed for easy handling purpose
- Levers , crossbars , handwheels are places for a mechanical advantage.